

Will Wu

Senior Backend Engineer

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SUMMARY

Senior backend engineer with 4+ years shipping and operating production systems in Go, Python, and TypeScript. Cut MongoDB p95 by 80% on a platform serving ~3M weekly API calls, root-caused a multi-layer SAML SSO failure down to a vendor XSD-validation conflict and shipped the production hot-patch, and built 3-layer distributed tracing across FastAPI and RabbitMQ. I take systems end to end: architecture, migration, and the incident afterwards.

WORK EXPERIENCE

APMIC — Sr. Backend Engineer

Nov 2025 – Present

Taipei, Taiwan · Python (FastAPI), React, Next.js, PostgreSQL, Hasura, RabbitMQ, Google ADK

Enterprise AI platform for on-premises LLM deployment and model fine-tuning; deliver end-to-end LLM / agent integrations and the full-stack enterprise systems around them for clients.

- **Delivered SAML 2.0 SSO and a privilege-isolating RBAC redesign** for an enterprise customer-service platform (iframe-embedded for a global insurance client) — root-caused a multi-layer SSO failure down to a vendor `xsi:type` XSD-validation conflict and shipped a production hot-patch; introduced a type discriminator to cleanly separate JIT-provisioned end-customers from operators, closing a privilege-escalation gap.
- **Built a real-time webchat agent-state system** — a 5-state machine (ready / not-ready / serving / busy / offline) enforced by PostgreSQL triggers, supervisor 1-to-many assignment, and cross-system mutual exclusion against the client's Cisco Finesse telephony platform (fail-open REST gate), covered by unit + Playwright e2e suites.
- **Self-built 3-layer distributed tracing** — contextvar `trace_id` propagated across FastAPI middleware and RabbitMQ message headers, with dual-sink structured logging and PII-safe business-event instrumentation across 6 modules (chosen over OpenTelemetry where no downstream collector existed).
- **Built automated web-scraping pipelines and a RAG knowledge-base integrity toolchain** — manifest-diff verification with exponential-backoff retry that surfaced 64 missing upstream documents and cut actionable gaps from 64 to 1, keeping the LLM retrieval corpus fresh and complete.

Vocus — Sr. Backend Engineer

Mar 2025 – Oct 2025

Taipei, Taiwan · Go (Gin), MongoDB, Redis, Qdrant, Cloudflare, AWS, GCP

- **Contributed to DDD refactoring of Taiwan's top-tier content-creator platform** (CDN-fronted, 94%+ Cloudflare cache-hit; backend handled ~3M weekly API calls for memberships, subscriptions, comments, recommendations, payments) — migrated legacy handlers into a 4-layer architecture across 73+ API modules; designed composition patterns (single-responsibility services + functional decorator chains with concurrent fan-out) to cut god-function complexity.
- **Owned end-to-end migration of the NewebPay payment integration** from server-side to iframe-redirect mode across all 4 payment types (subscription / one-time / recurring / donation) — moved card-data collection to the gateway's iframe to reduce PCI-DSS scope; redesigned token lifecycle across legacy + new paths.
- **Reduced MongoDB p95 latency on hotspot endpoints by 80%** (platform-wide API p95 80 ms, p50 4 ms) — restructured queries from `$lookup` cross-collection joins to single-collection reads + application-layer joins (goroutine fan-out), leveraging the document model instead of mimicking SQL.
- **Implemented batched async pipelines** — AWS SQS delayed-notification queue (batch 100–500, group-by aggregation, retry with backoff), hourly stats aggregation via worker pool (InsertMany replacing per-event writes), and periodic Qdrant-driven recommendation computation for article internal linking (SEO).
- **Migrated the recommendation system from BigQuery to self-hosted Qdrant vector DB, lowering infra cost by 20%** — eliminated per-query usage-based pricing for the highest-traffic query path.

ChengChi Tech — Sr. Backend Engineer

Apr 2024 – Mar 2025

New Taipei, Taiwan · Python (FastAPI), Next.js, Tauri, PostgreSQL, Redis, GCP, Azure

- Joined as the company's first and only Sr. backend engineer in an early-stage startup; architected and built 9 backend microservices from scratch (unified Python 3.12 / FastAPI / SQLAlchemy 2) covering cloud POS, short-term EV rental, AI meeting summarization, and multiple vertical ERPs.
- Owned all backend architectural decisions, technical selection, and code review; mentored junior engineers as the team grew from 4 to 7–8 people.
- Built development infrastructure from zero — self-hosted GitLab, CI/CD (GitLab CI + Ansible) reducing deployment time by 50%, IaC (Terraform) reducing infrastructure overhead by 30%, SonarQube quality gates.
- Shipped a commercial AI meeting-summarization product solo in 3 months — Whisperx (ASR with speaker diarization) + Ollama (local LLM) + a Tauri desktop app (Next.js 16 / React 19); currently serving paying customers.
- Shipped a cloud POS (F&B + retail) and a short-term EV-rental system (IoT + payment + SMS) — each ~2 months greenfield solo development.

AI-Rider — Backend Engineer

Mar 2022 – Nov 2023

New Taipei, Taiwan · Node.js (Koa), Python (Flask), MariaDB, Redis, RabbitMQ, Modbus TCP

- **Primary backend engineer on a POS system powering ~1,000 on-premise parking sites** (DuDuParking as flagship client, 50%+ of company revenue) — handled multi-tenant billing (time-tiered, monthly pass, EV-charging pre-auth, municipal contract rates).
- **Architected a hardware abstraction layer (ioserver) supporting 10+ device types** via Modbus TCP — ANPR cameras, gate arms, ETag, LED boards — with OOP design and a unified REST API hiding protocol complexity from upstream services.
- **Rebuilt the LED broadcasting system controlling thousands of LED panels across ~1,000 sites** — consolidated a 2-service legacy architecture (Go datasync + Python device backend) into a unified Turbo monorepo (Next.js + Electron + Python Flask), retiring the legacy Go service entirely.
- Optimized SQL queries reducing execution time by 50% for a customer with ~4–5M parking records (EXPLAIN analysis + compound indexes).
- Built a YouTube-based Asian facial dataset pipeline processing 1,000+ videos (multi-stream threading pool, incremental validation) to address facial-recognition model bias.

SIDE PROJECTS

bfxfundingbot — Algorithmic Crypto Funding Bot (Solo) 2026 · Python (asyncio), PostgreSQL, Docker, GitHub Actions · [case study](#)

- **Built and run a live, real-money Bitfinex margin-funding bot** end-to-end — event-sourced ledger with a periodic-reconcile correctness backbone (REST snapshot as source of truth, WebSocket as best-effort latency optimization, à la FIX drop-copy), single-writer exposure reconciliation, and balance-aware deployment gating; diagnosed and fixed 6+ production incidents on live capital, each shipped with a canary verification step.
- **Designed a quantitative strategy-validation framework** — walk-forward optimization, out-of-sample profitability with bootstrap confidence intervals and deflated Sharpe, and a bot-vs-idle attribution model.
- **Took the live canary from stalled to measured returns** — order-book data availability 3.9% → 99.6%, cold-start quoting 60 min → 14–27 s; realized net APR of 3.75–5.14% in lending weeks, reconciled against the exchange’s own ledger to within 0.1–2.9%.

divego — Full-Stack Migration (Solo) 2026 · TypeScript (Next.js, Hono, Drizzle) · [case study](#)

- **Migrated a FastAPI backend to a unified Next.js full-stack app** (Hono + Drizzle on Next.js 16) using the strangler-fig pattern — domain-by-domain vertical slices with a cross-stack unified error protocol, retiring the Python backend entirely (~567 tests).

SKILLS & TECHNICAL COMPETENCIES

Languages	Python, Go, TypeScript, Dart, Java
Backend	FastAPI, Gin, Koa, Flask, Spring Boot, Hono
Frontend / Desktop	Next.js, React, Tauri, Flutter
Databases	PostgreSQL, MariaDB, MongoDB, Redis, Qdrant (vector DB)
Data / Migration	SQLAlchemy, Drizzle, Atlas, Alembic, Flyway
Messaging	RabbitMQ, MQTT
AI / ML	Whisperx (ASR), Ollama (local LLM), Google ADK, PyTorch / CUDA
Cloud	GCP, AWS, Azure, Cloudflare, Alibaba Cloud
DevOps / Infra	Docker, k3s, GitLab CI, Terraform, Ansible, Self-hosted GitLab
Protocols	REST, gRPC, GraphQL (Hasura), SAML / OAuth, Modbus TCP, JSON-RPC, Webhooks

EDUCATION

MS, Data Analytics Engineering — George Mason University	2019 – 2021
BS, Chemical Engineering — Tatung University	2013 – 2016